SHORT COMMERCIAL DECK

CAMBRIDGE GRADES 1-9

AI Learning Platform

Our understanding of the problem and proposed solution

UNDERSTANDING THE SCOPE

What This Platform Must Deliver

FOR STUDENTS

After-hours support, objective-based study, AI help, practice, quizzes, and progress visibility.

FOR THE SCHOOL

A curriculum-aligned platform with teacher and admin oversight inside the school.

FOR THE CLIENT

A reusable product foundation that can later be sold to other schools.

This means the project is a **product platform**, not a single feature:

- Curriculum structure must exist behind the content
- Grades 1-6 and 7-9 cannot use the same learning experience
- Lessons need objective-level teaching, practice, and mastery tracking
- Teacher and admin workflows are part of the product, not optional extras

THE CHALLENGE

Why this is complex

The difficulty is the combination of curriculum, pedagogy, product, and platform.

Curriculum Complexity

Cambridge Primary and Lower Secondary, subjects, stages, topics, and learning objectives.

Student Experience

Grade 1 and Grade 9 need different wording, guidance, layout, and support style.

Content Complexity

Textbooks, lessons, examples, quizzes, source grounding, and curriculum mapping.

Platform Complexity

Students, teachers, admins, reporting, content management, and multi-school support.

ARCHITECTURE

How we structure the solution

Separating reusable platform layers to solve complexity.

EXPERIENCE LAYER

Student | Teacher | Admin

LEARNING LAYER

Objective pages · AI explanations · Examples · Practice · Quizzes · Mastery

CURRICULUM LAYER

Programme › Stage › Grade › Subject › Topic › Learning Objective

PLATFORM LAYER

Users · Content · Reporting · Settings · School Setup

KEY PRINCIPLES

- Curriculum first
- Stage-aware UX
- Objective-based delivery
- Reusable school product

PLATFORM ARCHITECTURE

System Modules

1 Platform Core
- School setup
- Users, roles, permissions
- Login and security
- Notifications, audit logs

2 Curriculum Management
- Stages and grades
- Subjects
- Topics and objectives
- Lesson mapping

3 Content Management
- PDF upload and extraction
- Lesson content and examples
- Practice and quiz content
- Media library

4 AI Tutor
- AI chat and explanations
- Examples and hints
- Step-by-step help
- Translation, voice/avatar

5 Student Learning
- Dashboard and topic view
- Objective and lesson pages
- Practice and quiz screens
- Notes and continue flow

6 Assessment and Mastery
- Question bank and quizzes
- Answer evaluation and feedback
- Mastery and progress tracking
- Weak area detection

7 Personalization
- Learner profile and diagnostics
- Difficulty and support adjustment
- Recommended lessons
- Stage-based experience rules

8 Teacher Module
- Dashboard and class management
- Student monitoring
- Content approval and notes
- Alerts and interventions

9 School Admin
- School and grade setup
- Teacher and student enrollment
- User administration
- Admin reporting

10 Reporting and Analytics
- Usage and activity reports
- Completion and progress
- Weak topic analysis
- AI usage reports

11 Engagement
- Points, stars, and badges
- Achievements and rewards
- Motivation messages
- Milestones

12 Accessibility and Language
- Multi-language and RTL/LTR
- Translation
- Text-to-speech
- Accessibility controls

AI INFRASTRUCTURE

Technology Requirements

This platform demands a deep, multi-layer AI stack — not a simple API integration.

LLM Core

Large Language Model

- Curriculum-aware reasoning
- Lesson explanation generation
- Age-adapted language output
- Answer evaluation and feedback
- Multi-language and translation

STT

Speech-to-Text captures student voice input for tutoring, questions, and dictation in real time.

TTS

Text-to-Speech reads lessons, explanations, and hints aloud — essential for younger grades and accessibility.

Voice AI

Conversational voice interface lets students talk to the AI tutor naturally, with context-aware responses.

Realtime Avatar

A real-time AI avatar that speaks, listens, and responds to students — delivering live interactive tutoring face-to-face.

AI ANALYSIS SKILLS

Performance Analysis
Tracks scores, time, and patterns per objective

Weak Area Detection
Identifies struggling topics automatically

Difficulty Calibration
Adjusts content complexity per student

Content Extraction
Parses and structures textbook material

Mastery Prediction
Forecasts readiness for next objective

Revision Recommendation
Suggests what to revisit before assessments

Every AI capability is grounded in the curriculum structure — responses, hints, and analysis are always tied to grades, subjects, topics, and learning objectives.

CONTENT PIPELINE

Data Ingestion & Processing

Transforms textbooks and school materials into structured curriculum-linked content, ready for AI tutoring, lesson delivery, practice, quizzes, and progress tracking.

Upload
- Textbooks
- PDF documents
- School materials
- Reference files

→

Extract
- Text and structure
- Images and diagrams
- Tables and formulas
- Section headings

→

Organize
- Lessons and examples
- Practice questions
- Quiz items
- Source references

→

Map
- Grade and stage
- Subject and topic
- Learning objectives
- Curriculum tags

→

Structured Content
- Curriculum-linked
- AI-ready
- Searchable
- Reusable

STRUCTURED CONTENT POWERS

- AI Tutoring
- Lesson Delivery
- Practice
- Quizzes
- Progress Tracking

MOVING FORWARD

Next Steps

Four actions to move from understanding to delivery.

01

Confirm Scope & Priorities

Agree on which modules and features are in Phase 1 and what can be phased to later deliveries.

02

Curriculum Data Handoff

Share existing Cambridge syllabus materials, PDFs, and textbooks so ingestion pipeline scoping can begin.

03

Technical Discovery Session

A focused session to map existing systems, integrations, user roles, and any constraints that affect architecture.

04

Proposal & Timeline

Detailed project proposal with phased timeline, deliverables per phase, and agreed milestones.